

Learning Manipulation

Jitendra Malik

Learning Fine-Grained Bimanual Manipulation with Low-Cost Hardware

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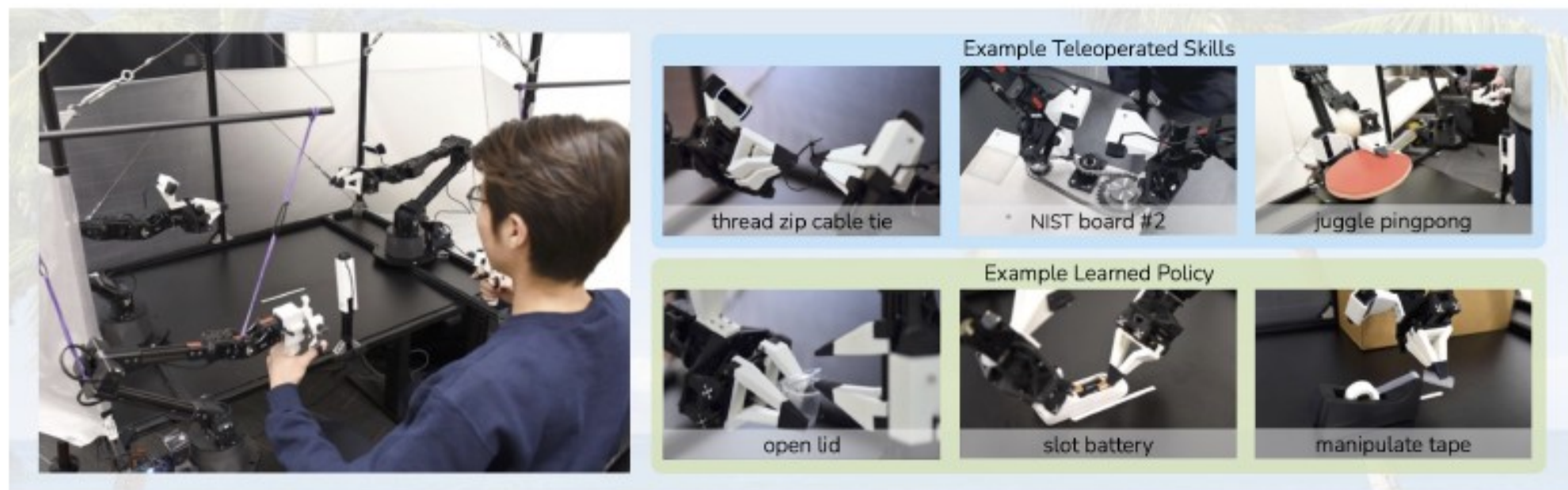
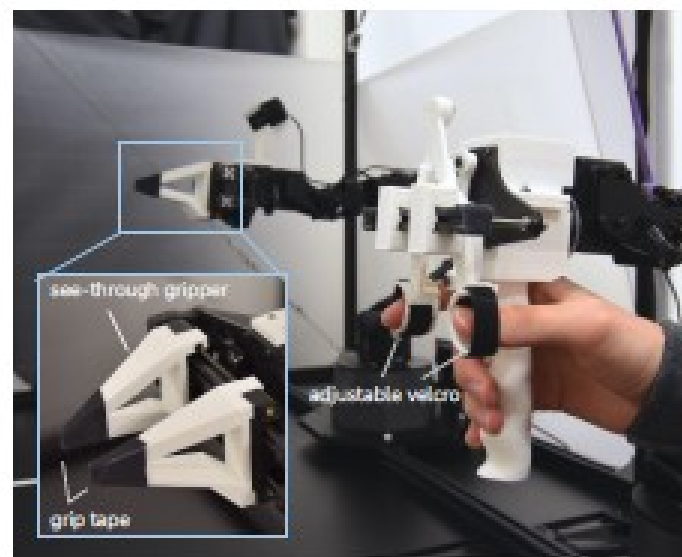
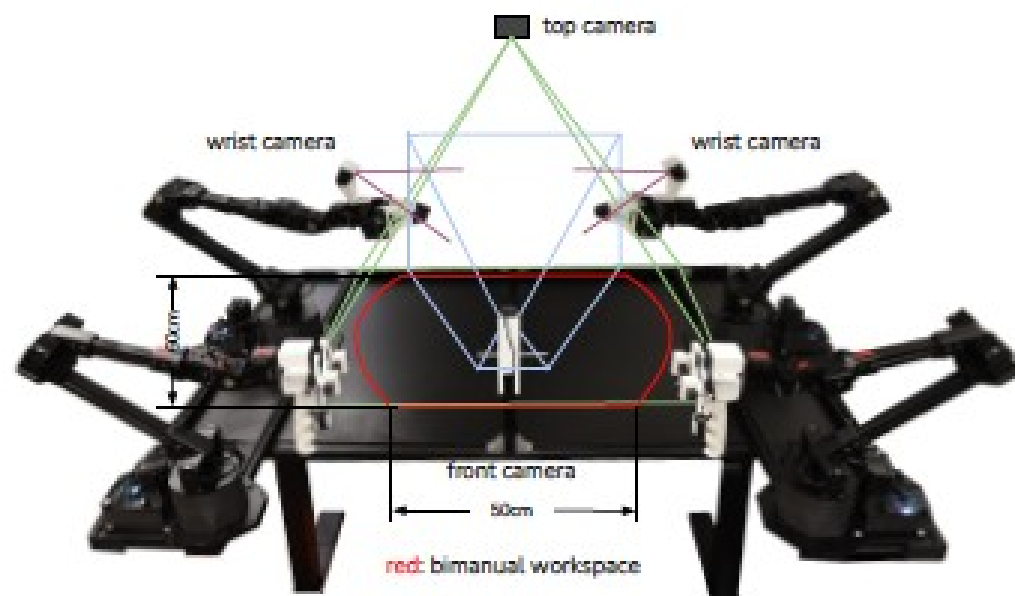


Fig. 1: ALOHA 🍷 : A Low-cost Open-source Hardware System for Bimanual Teleoperation. The whole system costs <\$20k with off-the-shelf robots and 3D printed components. *Left*: The user teleoperates by backdriving the leader robots, with the follower robots mirroring the motion. *Right*: ALOHA is capable of precise, contact-rich, and dynamic tasks. We show examples of both teleoperated and learned skills.



ViperX 6dof Arm (follower)

#Dofs	6+gripper
Reach	750mm
Span	1500mm
Repeatability	1mm
Accuracy	5-8mm
Working Payload	750g

Fig. 3: *Left*: Camera viewpoints of the front, top, and two wrist cameras, together with an illustration of the bimanual workspace of *ALOHA*. *Middle*: Detailed view of the “handle and scissor” mechanism and custom grippers. *Right*: Technical spec of the ViperX 6dof robot [1].

Precise tasks			Contact-rich tasks			Dynamic tasks		
Thread zip tie (in 5s)	Pick credit card (in 5s)	Open ziploc (in 5s)	RAM insertion (in 30s)	Turn pages (in 30s)	NIST board belt (in 30s)	Juggle ping pong (for 5 times)	Balance ping pong (for 20s)	Swing open bags (in 5s)
10/10	10/10	10/10	9/10	10/10	8/10	4/10	9/10	10/10

TABLE I: *Teleoperation success.* We report the number of successes out of 10 trials. The task is only successful if completed within the time limit specified in the table, excepting the two ping pong tasks where success is measured in duration. All objects used are off-the-shelf.

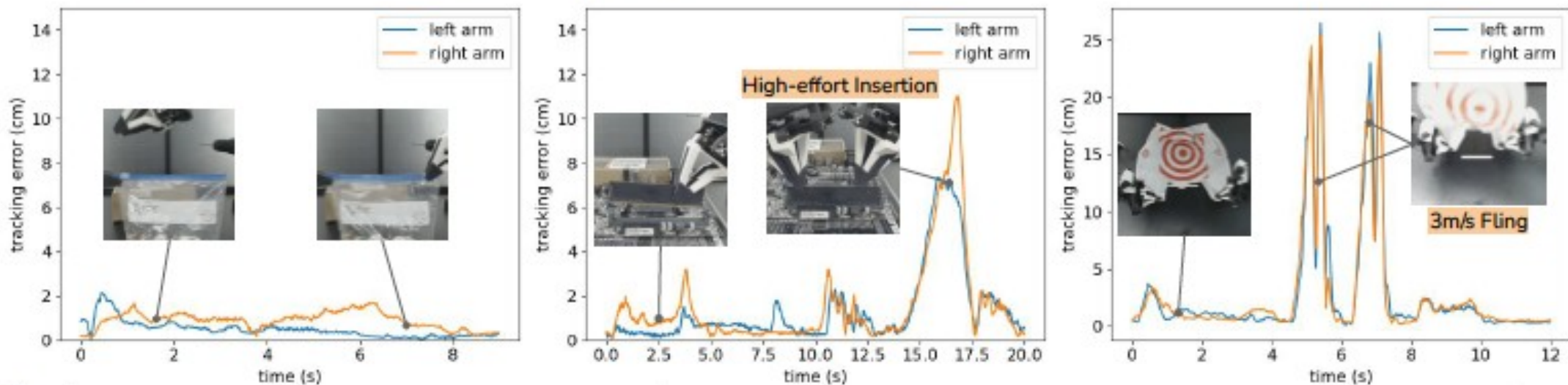


Fig. 4: *Teleoperation accuracy.* We plot the position tracking error of the *ALOHA* teleoperation system for three representative tasks. (a) A relatively smooth task of sliding open a ziploc bag. The tracking error is consistently below 2cm with an average of 0.68cm, comparable to the accuracy of the robot arm. (b) The tracking error can spike when the follower arms are exerting high-amount of force, e.g. inserting a RAM stick into a tight-fitting slot. This is due to the non-bilateral nature of the system. (c) High tracking error can also occur during sudden acceleration, as in the case of swinging open a plastic bag. This is due to a combination of teleoperation delay and motor torque limitations.

ALOHA Unleashed: A Simple Recipe for Robot Dexterity

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Abstract: Recent work has shown promising results for learning end-to-end robot policies using imitation learning. In this work we address the question of how far can we push imitation learning for challenging dexterous manipulation tasks. We show that a simple recipe of large scale data collection on the ALOHA 2 platform, combined with expressive models such as Diffusion Policies, can be effective in learning challenging bimanual manipulation tasks involving deformable objects and complex contact rich dynamics. We demonstrate our recipe on 5 challenging real-world and 3 simulated tasks and demonstrate improved performance over state-of-the-art baselines. The project website and videos can be found at aloha-unleashed.github.io.

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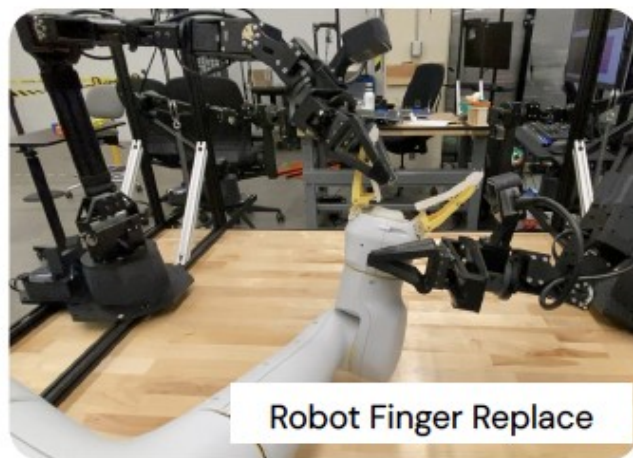


Figure 1: *ALOHA Unleashed* trains a transformer encoder-decoder architecture with a diffusion loss to learn highly dexterous bimanual manipulation tasks like hanging a shirt and tying shoelaces.

- **Shirt hanging (Shirt):** This task requires hanging a shirt on a hanger. The detailed steps include flattening the shirt, picking a hanger off a rack, performing a handover, picking up the shirt, precisely inserting both sides of the hanger into the shirt collar, then hanging the shirt back on the rack. This is a challenging task that requires multiple steps with deformable manipulation, insertions, and dexterous pick and placing behaviors like hooking and unhooking the hanger from the rack. We construct two variants of this task: **ShirtEasy** has a more constrained initialization, with the shirt flattened and centered on the table. **ShirtMessy** allows the initialization of the shirt to be rotated and crumpled and has significantly more variance in starting location.
- **Shoelace tying (Lace):** This task requires centering a shoe on the table, straightening the laces, then performing a maneuver to tie the laces in a bow. We construct two variants of this task: **LaceEasy** has a constrained initialization with the shoe centered on the table and laces extended outward. **LaceMessy** allows ± 45 degree variance in the angle of the shoe and does not require the laces to be flattened.
- **Robot finger replacement (FingerReplace):** This task requires removing a robot finger from a slotted mechanism, picking the replacement finger, reorienting the finger, then performing a precise insertion back into the slot with millimeter tolerance.
- **Gear insertion (GearInsert):** This task requires inserting 3 plastic gears onto a socket with millimeter precision with a friction fit, while ensuring that the gear is fully seated and the teeth on the gear mesh with neighboring gears.
- **Random kitchen stack (RandomKitchen):** This task requires cleaning up a randomly initialized table by stacking bowls, cups, and utensils and placing the stack at the center of the table.

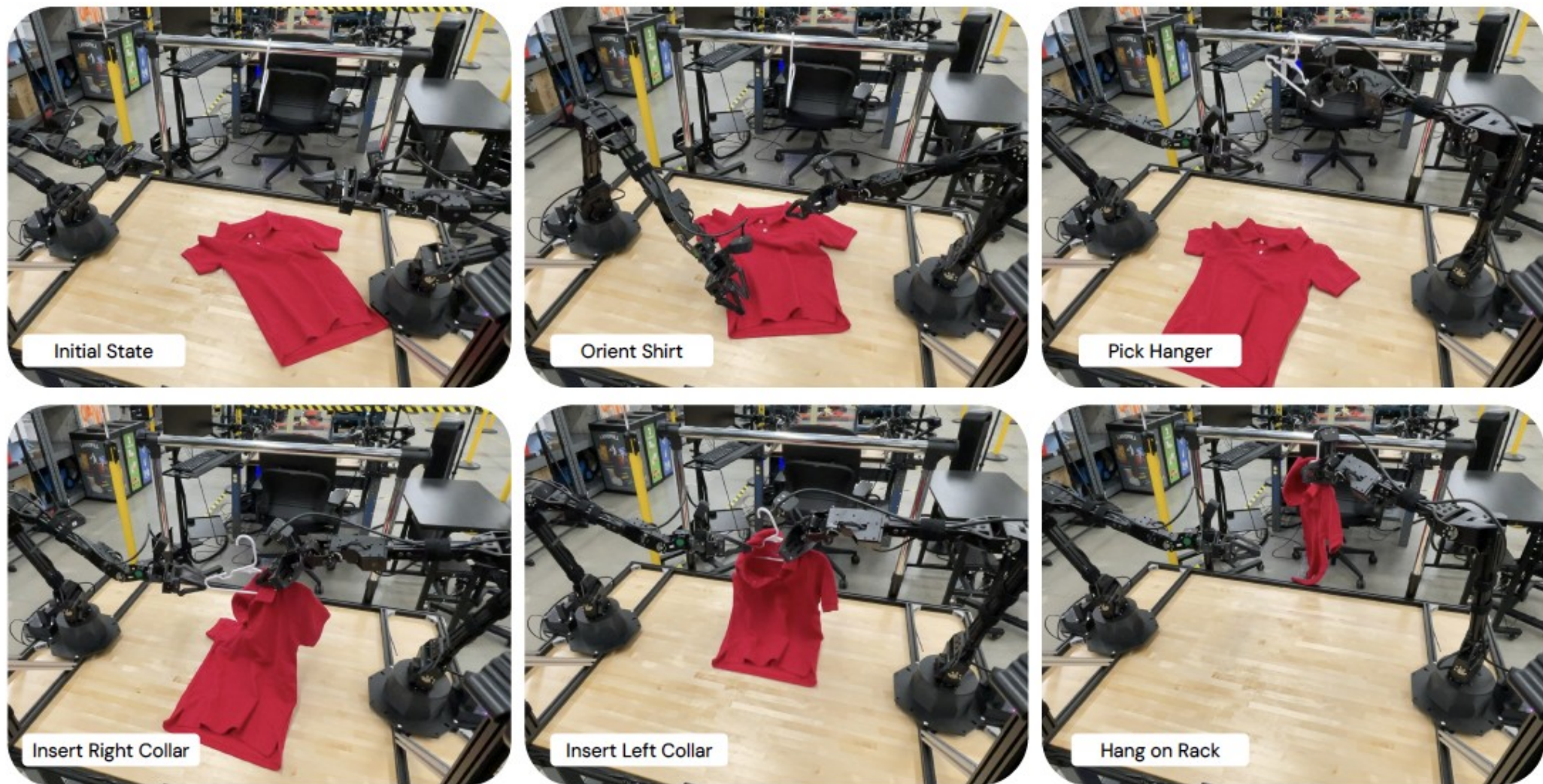


Figure 3: *Shirt Episode*. This task is long horizon, involves a deformable object, and requires several dexterous bimanual behaviors to achieve the end goal of hanging the shirt on the rack.

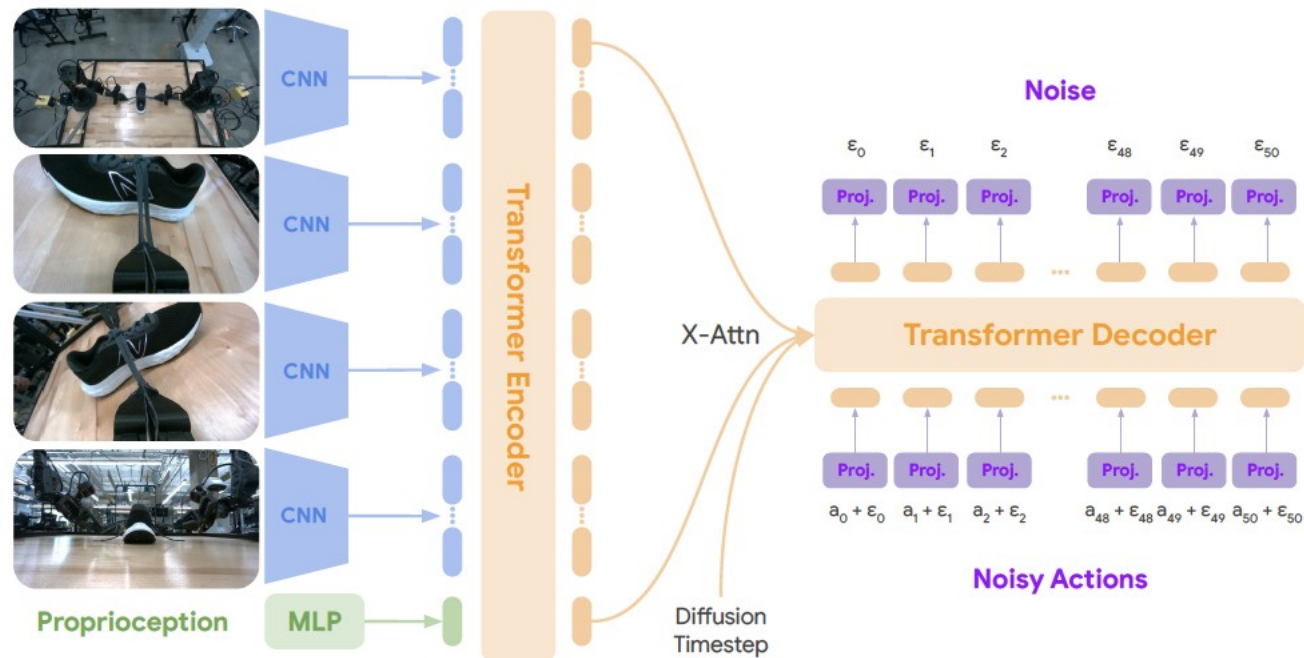


Figure 2: **Policy Architecture.** Each camera view is embedded with a ResNet50 [46]. The Transformer Encoder embeds the observations and produces latent embeddings. The Transformer Decoder takes in noisy actions and cross-attends to the latent embeddings produced by the encoder, outputting the predicted noise. The decoder portion runs 50 times during inference to iteratively denoise an action chunk.

Task	Success Rate	Number of Demonstrations
ShirtEasy	75%	8658 (5345 Easy; 3313 Messy)
ShirtMessy	70%	
LaceEasy	70%	5133 (2212 Easy; 2921 Messy)
LaceMessy	40%	
FingerReplace	75%	5247
GearInsert-1	95%	4005
GearInsert-2	75%	
GearInsert-3	40%	
RandomKitchen-Bowl	95%	3198 (216 In-Domain)
RandomKitchen-Bowl+Cup	65%	
RandomKitchen-Bowl+Cup+Fork	25%	

Table 1: Success rates and number of demonstrations for 5 real tasks, with separate models for each task. Shirt and Lace tasks each have a single model trained on 8658 and 5133 demonstrations, respectively, and are separately evaluated on Easy and Messy variants of the tasks. The GearInsert and RandomKitchen model evaluations are broken down by total progress on the task.

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Google DeepMind



[Paper](#)

Abstract

Physical Intelligence (π)

π_0 : Our First Generalist Policy

Published October 31, 2024
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Paper [\$\pi_0\$.pdf](#)



Local Policies Enable Zero-shot Long-horizon Manipulation

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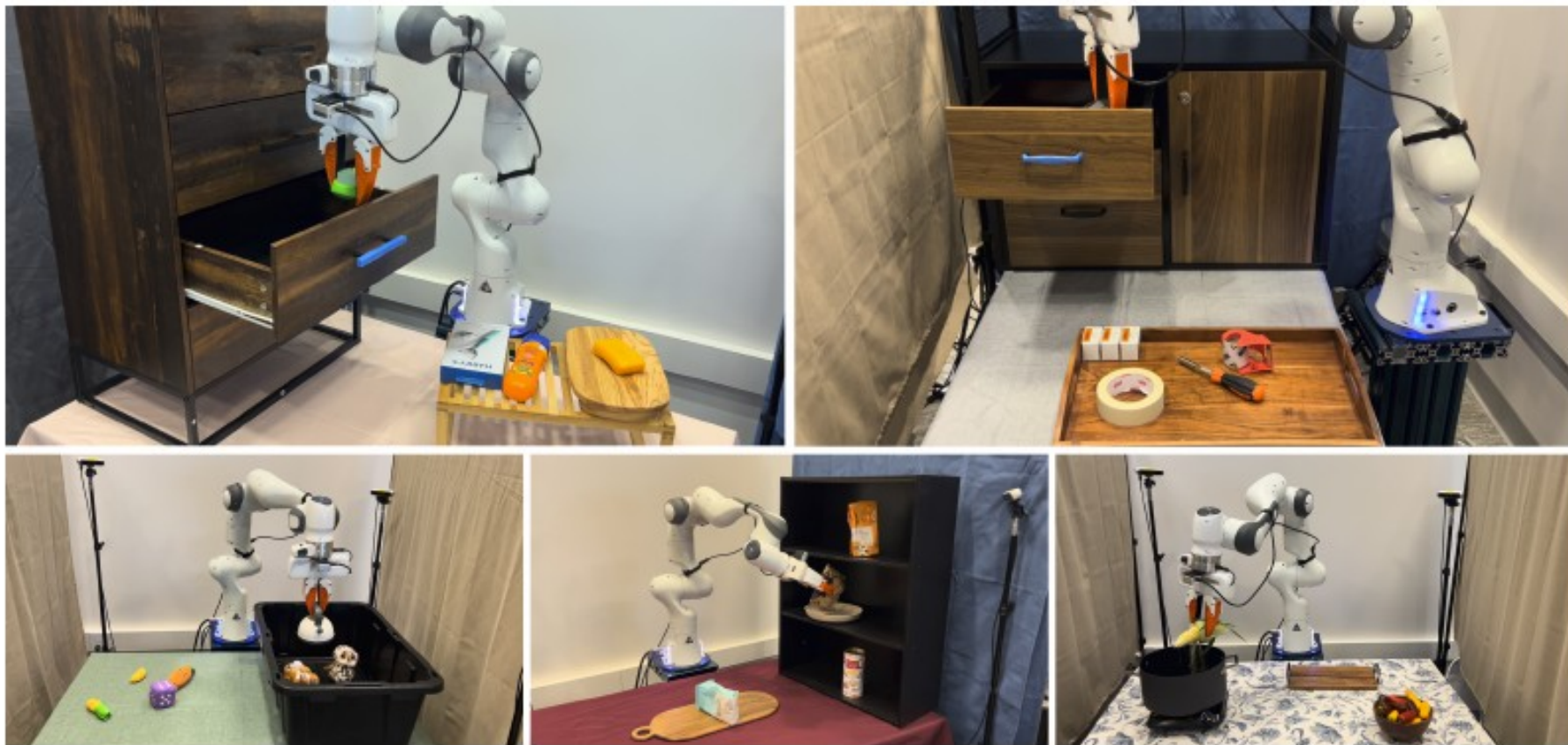


Fig. 1: Zero-shot Long-horizon Manipulation Our approach trains a library of generalist manipulation skills in simulation and transfers them zero-shot to long-horizon manipulation tasks. We show a single, text-conditioned agent can manipulate unseen objects, in arbitrary poses and scene configurations, across long-horizons in the real world, solving challenging manipulation tasks with complex obstacles.

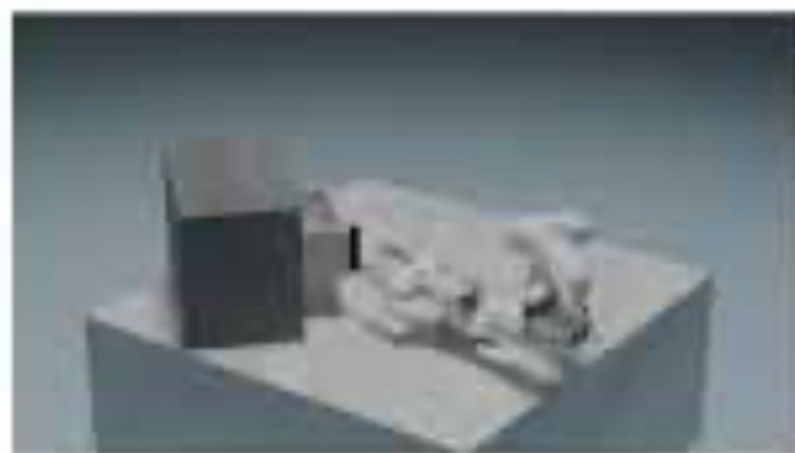
Local Policy Skill Library



Pick



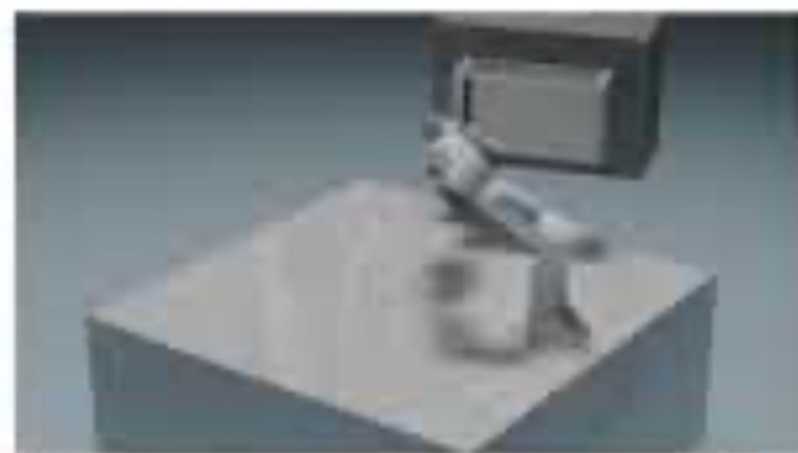
Place



Grasp Handle



Open



Close

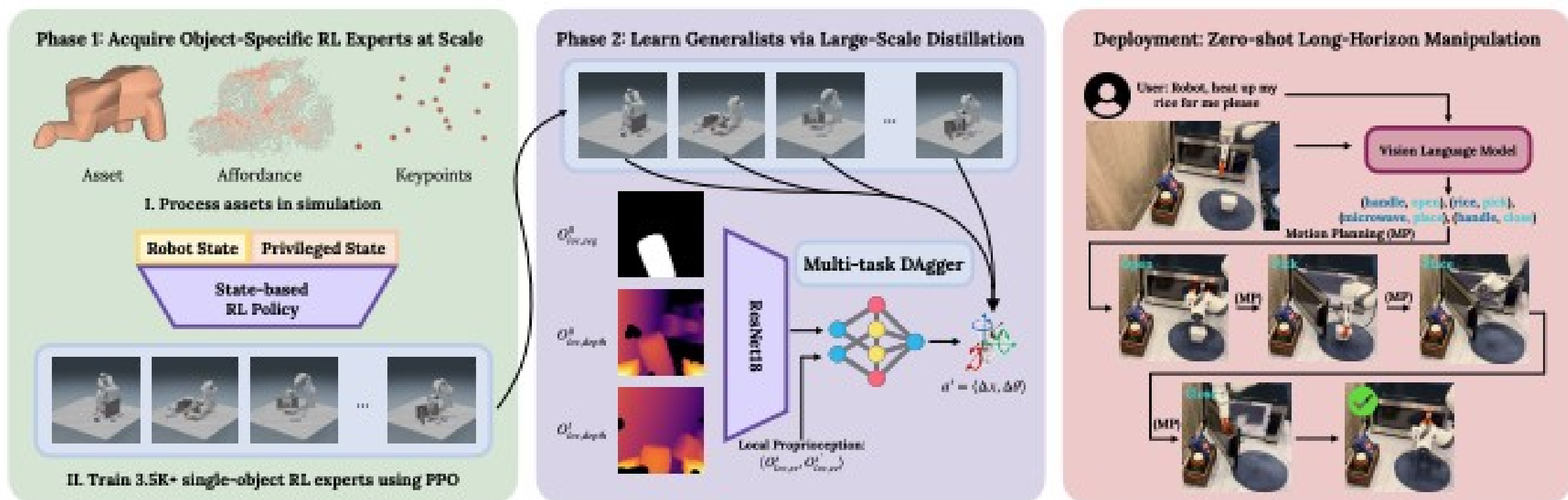


Fig. 3: ManipGen Method Overview (*left*) Train 1000s of RL experts in simulation using PPO (*middle*) Distill single-task RL experts into generalist visuomotor policies via DAgger (*right*) Text-conditioned long-horizon manipulation via task decomposition (VLM), pose estimation and goal reaching (Motion Planning) and sim2real transfer of local policies

	Cook	Replace	CabinetStore	DrawerStore	Tidy	Avg
<i>Stages</i>	2	4	4	6	8	4.8
OpenVLA	0% (0.1)	0 (0.0)	0% (0.0)	0 (0.0)	0 (0.0)	0% (.02)
SayCan	80% (1.7)	10% (1.3)	70% (3.5)	20% (3.6)	20% (4.8)	40% (3.0)
LLMTrajGen	70% (1.5)	0% (0.6)	0% (0.6)	0% (1.0)	0% (2.6)	14% (1.3)
VoxPoser	70% (1.4)	0% (0.8)	0% (0.8)	0% (0.9)	10% (4.4)	16% (1.7)
Ours	90% (1.9)	80% (3.7)	90% (3.9)	60% (4.7)	60% (7.2)	76% (4.3)

TABLE III: Zero-shot Long Horizon Manipulation We report task success rate and average number of stages completed per real world task. ManipGen outperforms all methods on each task, achieving 76% with 4.28/4.8 stages completed on average.